efrei
INGÉNIEURS - NUMÉRIQUE
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Analysis and Prediction of High-Traffic Recipes

Exploratory analysis and classification models

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Logistic Regression · Random Forest · Gradient Boosting

Outline

1. **Context and objective**
2. **Exploratory analysis** — cleaning · links
3. **Modeling** — 3 models · ROC
4. **Conclusion and limitations**

Guiding question

Which recipes should be featured to maximize traffic, and therefore subscription sales?

Understand the data → identify traffic drivers → predict.

Context and objective

The recipe website wants to sell subscriptions. The more visits a recipe attracts, the higher the chance of a sale.

Objective: predict which recipes will drive high traffic, in order to choose the ones to feature.

947
recipes

8
variables

10
categories

61%
high traffic



Exploratory analysis

Cleaning, distributions and links with traffic

Data and cleaning

1 Target

Empty values of `high_traffic` = "Low", then coded 0 / 1.

2 Categories

11 values in the data: "Chicken Breast" merged with "Chicken" to get the expected 10.

3 Servings

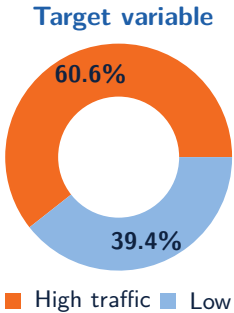
"4 as a snack" and "6 as a snack" converted to 4 and 6; column cast to integer.

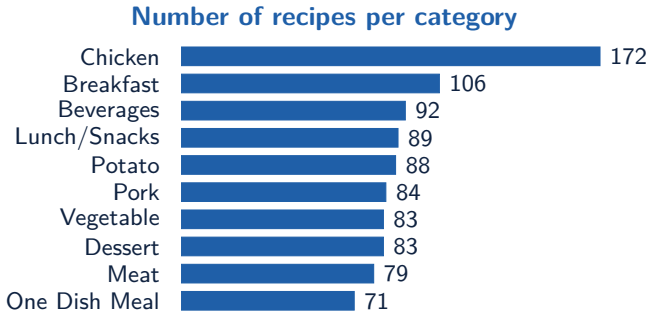
4 Missing values

52 recipes (5.5%) without nutrition data, imputed with their category median.

No duplicates; recipe identifier is unique.

Overview of the target and categories





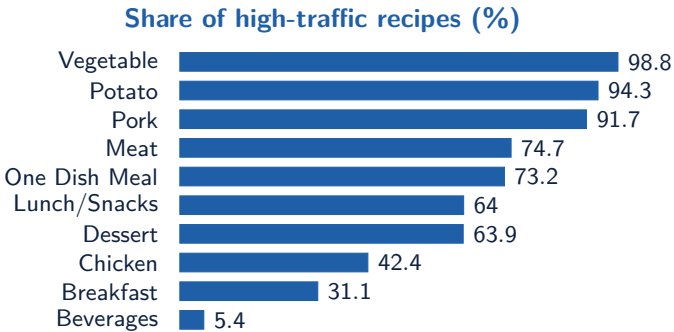
Moderately imbalanced classes (61 / 39) and well-represented categories (71 to 172 recipes).

Descriptive statistics

Variable	Mean	Median	Maximum	Skewness
Calories	431.1	296.9	3,633	2.11
Carbohydrates (g)	34.4	21.2	530	3.87
Sugar (g)	8.9	4.4	149	4.29
Protein (g)	23.9	11.2	363	3.59

Means well above medians and extreme values: strongly skewed distributions. `log1p` transform for the linear model; unnecessary for tree-based models.

Category explains most of it



0.58

Cramér's V

χ^2: $p < 0.001$

From 5% (beverages) to 99% (vegetables): category is by far the most discriminating variable.

Nutrition and traffic: weak effects

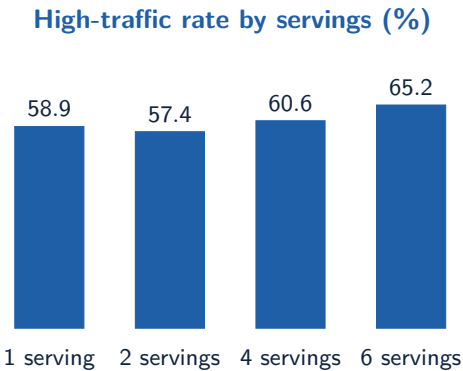
Mann-Whitney test (high vs low traffic)

Variable	U statistic	p-value
Calories	115,214	0.0472
Carbs	115,446	0.0412
Sugar	98,956	0.0491
Protein	**120,705**	**0.0009**

Reading

- Protein: the only clear link with the target.
- Calories, carbs, sugar: p close to 0.05, with no correction for multiple tests.
- Effect much weaker than that of category.

Servings and traffic: no link



$p = 0.43$

χ^2 test

Rates stay between 57% and 65%: the number of servings does not distinguish popular recipes.

Missing values: a false signal

High-traffic rate

59.8%
nutrition provided
(895 recipes)

75.0%
nutrition missing
(52 recipes)

The gap disappears within a category: missing values are concentrated in high-traffic categories (none in *Beverages* or *Breakfast*). The direction of the gap changes from one category to another.

Cross-validation test (Random Forest)

	F1	ROC AUC
With	0.7996	0.8137
Without	0.7910	0.8138

✓ **Variable removed: identical AUC, no measurable gain, simpler model.**

Summary of the exploration

1 **Target**	61% high traffic: moderately imbalanced classes, so F1 and AUC are preferred metrics.
2 **Category**	Most discriminating variable ($V = 0.58$, from 5% to 99%): kept, one-hot encoded.
3 **Nutrition**	Skewed distributions, weak effect: `log1p` for the linear model.
4 **Servings, missing**	No effect of their own: `nutrition_missing` dropped, servings kept at no cost.



Modeling

Logistic Regression, Random Forest and Gradient Boosting

Modeling approach

1 Split
80 / 20 stratified
757 train · 190 test

2 Preprocessing
Log + scaling (linear), one-hot

3 Cross-validation
5 stratified folds
F1, AUC, precision

4 Tuning
GridSearch on F1, 3 models

5 Final test
Evaluation on 190 recipes

All preprocessing lives inside scikit-learn pipelines: no data leakage between training and validation.

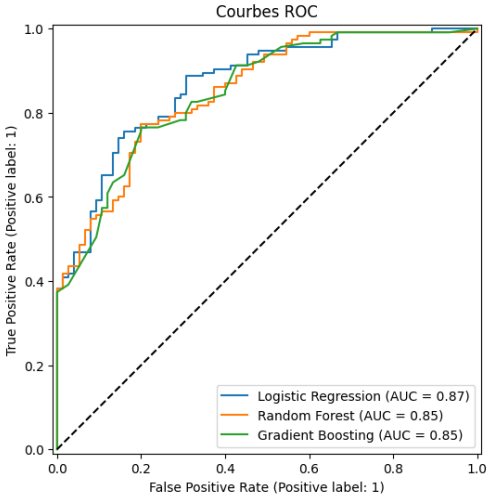
Three models, similar scores

Model	F1 (CV)	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	**0.803**	**0.779**	**0.848**	**0.774**	**0.809**	**0.869**
Random Forest	0.796	0.753	0.788	0.809	0.798	0.852
Gradient Boosting	0.801	0.779	0.854	0.765	0.807	0.850

CV = cross-validation on the training set; other columns = test set (190 recipes).

F1 differences below 0.02: the models are equivalent. Selected: logistic regression (best AUC, simplest and most interpretable).

ROC curves of the three models



ROC curves on the test set (190 recipes)

0.87

AUC of the selected model

The three curves are very close and clearly above the diagonal (chance).

The AUC is the probability that a high-traffic recipe gets a higher score than a low-traffic one.

The selected model in detail

Confusion matrix (test set, 190 recipes)

	Predicted Low	Predicted High
Actual Low	59	16
Actual High	26	89

0.85 Precision

When the model predicts high traffic, it is right 85 times out of 100.

0.77 Recall

It finds 77% of the high-traffic recipes.

0.87 ROC AUC

Good ability to rank recipes.

Decision threshold at 0.5.

Conclusion

Key takeaways

- Category is the dominant driver of traffic.
- Nutrition and servings add little.
- Logistic regression: F1 0.81 and AUC 0.87 on the test set.
- The three models are equivalent; the simplest is selected.

Limitations and next steps

- Small sample: 947 recipes, 190 in the test set.
- No data on ingredients or preparation time.
- Imputation computed before the train / test split: should move into the pipeline.
- Adjust the threshold to the cost of a wrongly promoted recipe.

Open the interactive app